

DeepResearch Agent

System Architecture, Process Workflow & Technical Documentation

Project Name:	DeepResearch Agent
System Type:	Autonomous Multi-Agent AI Research Engine
Execution Cycle:	Plan → Act → Observe → Evaluate → Synthesize
Target Deliverable:	Cited Research Reports & Formatted PDF Exports
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1. Executive Summary & Core Objectives (What is Wanted)

Problem Statement: Standard conversational AI interfaces fail at multi-layered academic or enterprise research due to single-turn prompt limitations, lack of live source attribution, and output restricted to unstructured raw text.

Core Solution: DeepResearch Agent provides an end-to-end autonomous research workflow that deconstructs user topics, queries live web indexes, extracts verified facts, evaluates knowledge sufficiency, and compiles publication-grade PDF reports complete with numerical citations.

Key Agent Capabilities:

- **Autonomous Planning:** Automatically splits complex prompts into targeted sub-questions.
- **Multi-Source Gathering:** Performs web and image searches with rate-limit handling.
- **Granular Fact Extraction:** Distills key facts from web pages before synthesis.
- **Iterative Quality Check:** Evaluates answer sufficiency and fills information gaps.
- **Cited Synthesis & PDF Export:** Produces structured reports with [1], [2] citations and PDF exports.

2. System Architecture & Tech Stack (What is Used Where)

The system is divided into a modern React frontend, a FastAPI asynchronous backend, and a LangGraph agentic decision graph.

Layer	Technology	Purpose & Module
Frontend UI	React 18 + Vite	User interface, state coordination (src/App.jsx)

Data Polling	TanStack Query	Polls /api/research/{id}/status every 2s
HTTP Client	Axios	API request dispatching (src/services/api.js)
Backend Framework	FastAPI	Async API routes & background tasks (app/main.py)
Agent State Graph	LangGraph	Cyclic execution workflow (app/agents/agent_graph.py)
LLM Integration	Ollama / OpenAI	Planning, evaluation & synthesis prompting
Web Search Tool	DuckDuckGo (ddgs)	Live web search tool (app/tools/web_search.py)
Image Acquisition	DDG Images + httpx	Image downloading (app/tools/image_search.py)
PDF Generator	ReportLab + Pillow	PDF compilation (app/services/pdf_generator.py)

3. End-to-End Execution Process (The Workflow)



Detailed Process Steps:

- 1. **Initiation:** User submits topic on frontend (`ResearchForm.jsx`). POST `/api/research/start` creates session and triggers background task `run\_research`.
- 2. **Planning (`plan\_research`):** LLM creates 3-5 targeted research questions to cover multiple angles of the user topic.
- 3. **Tool Search & Fact Extraction (`conduct\_research`):** System searches DuckDuckGo for each sub-question, extracts 2-3 key evidence facts per source snippet using LLM, and logs reasoning.
- 4. **Quality Evaluation & Gap Analysis (`evaluate\_research`):** Evaluates collected sources. If insufficient, generates gap queries via `research\_more` and loops back to research.
- 5. **Synthesis (`synthesize\_answer`):** Synthesizes evidence into a cited markdown response with inline source brackets ([1], [2]).

6. **PDF Compilation (`PDFReportGenerator`)**: On download, system fetches images via `ImageSearchTool`, constructs ReportLab layout (Cover, Metadata, Plan, Cited Findings, Image Gallery, References), and outputs PDF.

4. Concrete Example Dry Run Trace

Topic: *Impact of Quantum Computing on Financial Cybersecurity*

Step	Execution Details
1. Plan Generation	LLM outputs 3 sub-questions: 1. <i>How do quantum algorithms (Shor's) threaten RSA encryption in finance?</i> 2. <i>What Post-Quantum Cryptography (PQC) standards are banks adopting?</i> 3. <i>What is the estimated timeline for quantum cybersecurity readiness?</i>
2. Web Search & Fact Extraction	• Source [1]: 'Quantum Threats to Financial Systems' Fact: Shor's algorithm efficiently factors large integers, invalidating RSA-2048. • Source [2]: 'NIST PQC Standards' Fact: NIST finalized CRYSTALS-Kyber for post-quantum key encapsulation.
3. Evaluation	Evaluator checks total sources (6 found across 3 sub-questions). Determines quality is SUFFICIENT . Proceeds to synthesis.
4. Citation Synthesis	'Quantum computing threatens financial systems by breaking RSA encryption via Shor's algorithm [1]. Central banks are migrating to NIST-approved algorithms such as CRYSTALS-Kyber [2].'
5. PDF Output	Downloads 2 related graphics via `ImageSearchTool`, builds ReportLab document layout, and delivers research_report_eb719a42.pdf.

Documentation compiled successfully. File ready for distribution.